ROAD TRANSPORT AGENCY DASHBOARD

Project Description:

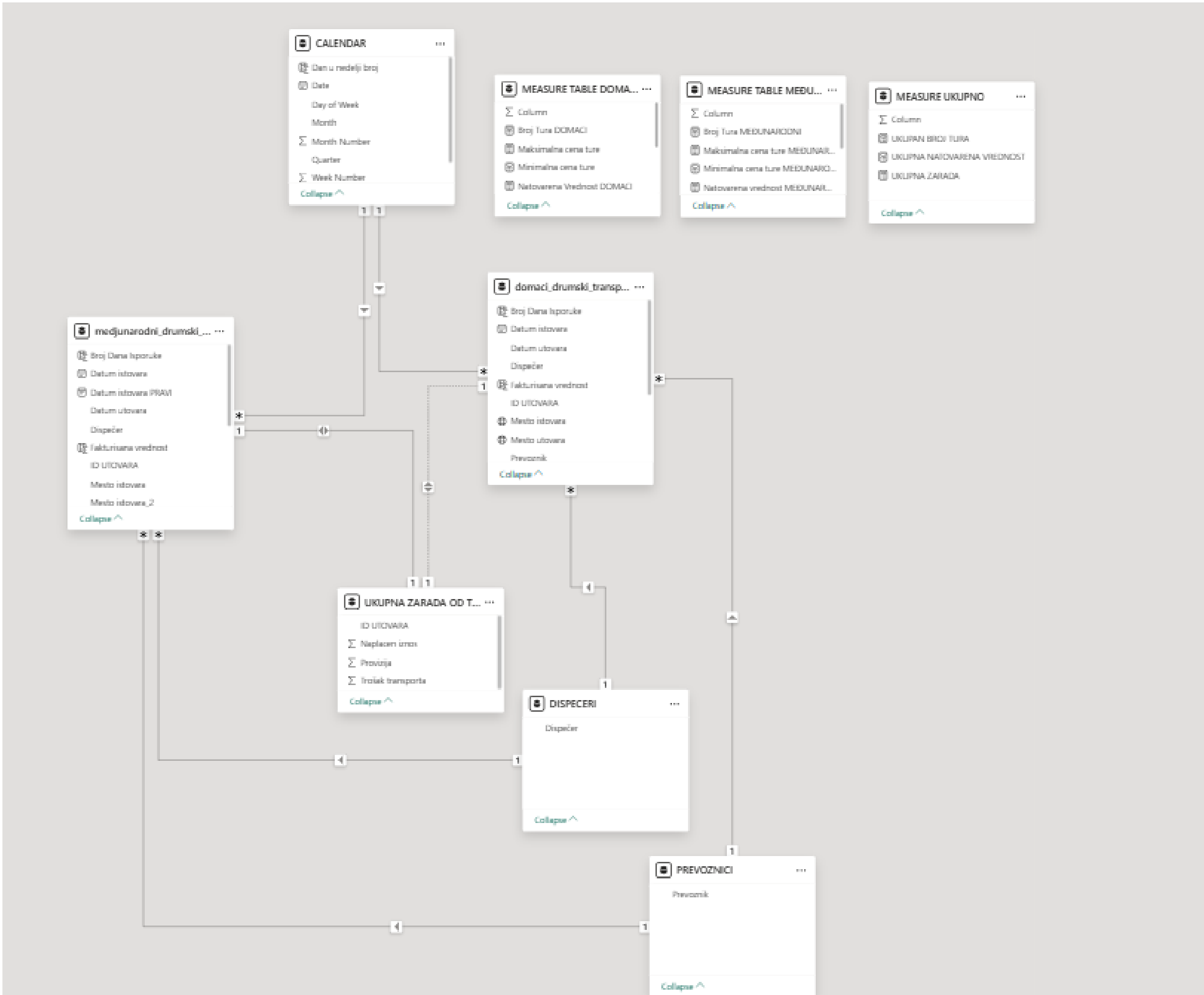
This project is centered around the analysis and reporting of transportation data, specifically focusing on both domestic and international road transport activities. The model integrates multiple fact and dimension tables to provide comprehensive insights into the performance, revenues, costs, and resources associated with transportation services. The key components of the data model are as follows:

Fact Tables:

1. **Medjunarodni Drumski (International Road Transport)**: This fact table contains data related to transportation activities involving international routes, including the associated costs, revenues, and other performance metrics.
2. **Domaci Drumski (Domestic Road Transport)**: This fact table focuses on domestic road transport activities, capturing similar metrics as the international transport but restricted to national routes.

Dimension Tables:

1. **Ukupna Zarada Od Transporta (Total Revenue from Transport)**: This dimension table holds detailed information on the revenues, costs, and provisions related to the transportation agency. It provides essential financial metrics to assess the profitability of the transport operations.
2. **Calendar Table**: This table enables time-based analysis, providing the structure necessary for aggregating and filtering data by dates, months, quarters, and years.
3. **Dispeceri (Dispatchers/Employees)**: This dimension table stores information about the employees involved in dispatching, including their roles and responsibilities in the transport operations. It allows for detailed employee-level performance tracking and analysis.
4. **Prevoznici (Carriers/Transport Partners)**: This table includes data about the transport partners, or carriers, whose trucks are engaged in transportation services. It helps in understanding the collaboration with external partners and their contribution to the overall transport operations.



Purpose of the Project:

The goal of this project is to create a unified data model that allows for detailed analysis and reporting of both domestic and international road transport operations. By leveraging the fact and dimension tables, stakeholders can gain insights into financial performance (revenue, costs, and profitability), operational efficiency (employee and partner performance), and temporal trends (seasonality and time-based patterns). This model supports decision-making processes, strategic planning, and operational improvements in the transportation business.

DASHBOARD STRUCTURE

Dashboard has a few parts:

1. Dashboard Content
2. Total Revenue of Road Transport Agency
3. Total Revenue of Domestic Road Transport
4. Loading and Unloading Locations in Domestic Transport
5. Types of Loaded Goods in Domestic Transport
6. Types of Trucks in Domestic Transport
7. Partners in Domestic Transport
8. Total Revenue of International Road Transport
9. Loading and Unloading Locations in International Transport
10. Types of Loaded Goods in International Transport
11. Types of Trucks in International Transport
12. Partners in International Transport

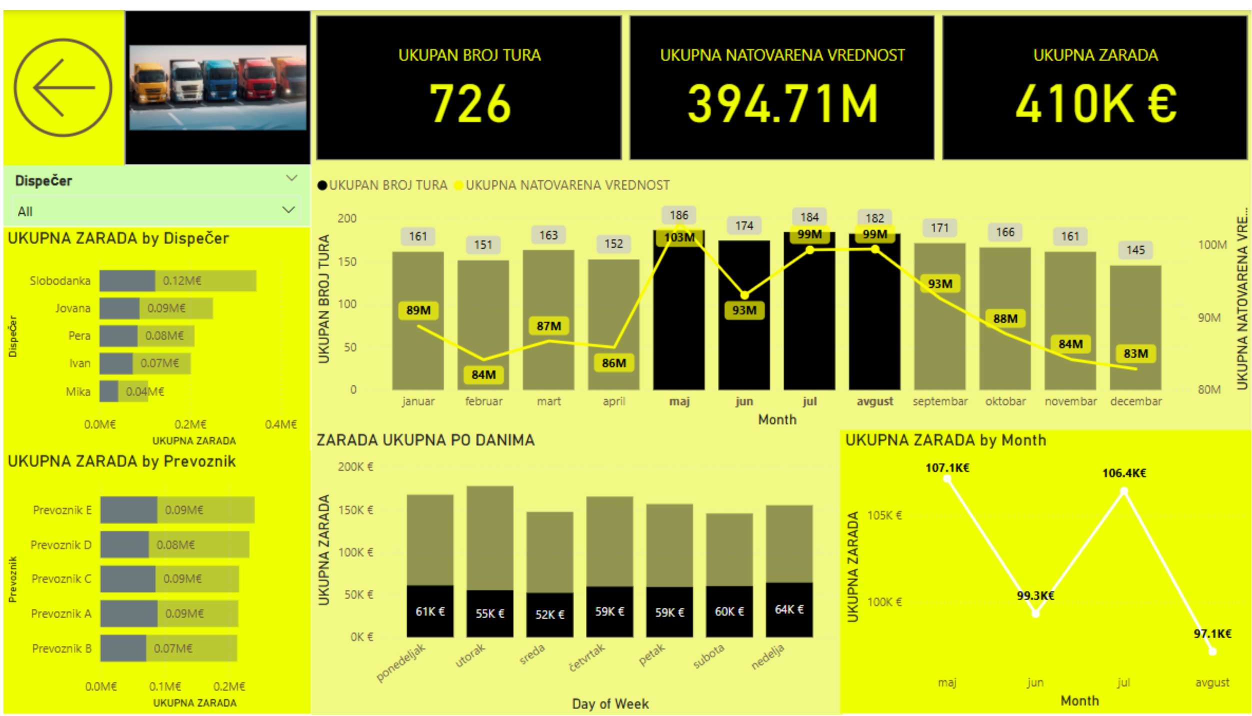
INSIGHTS

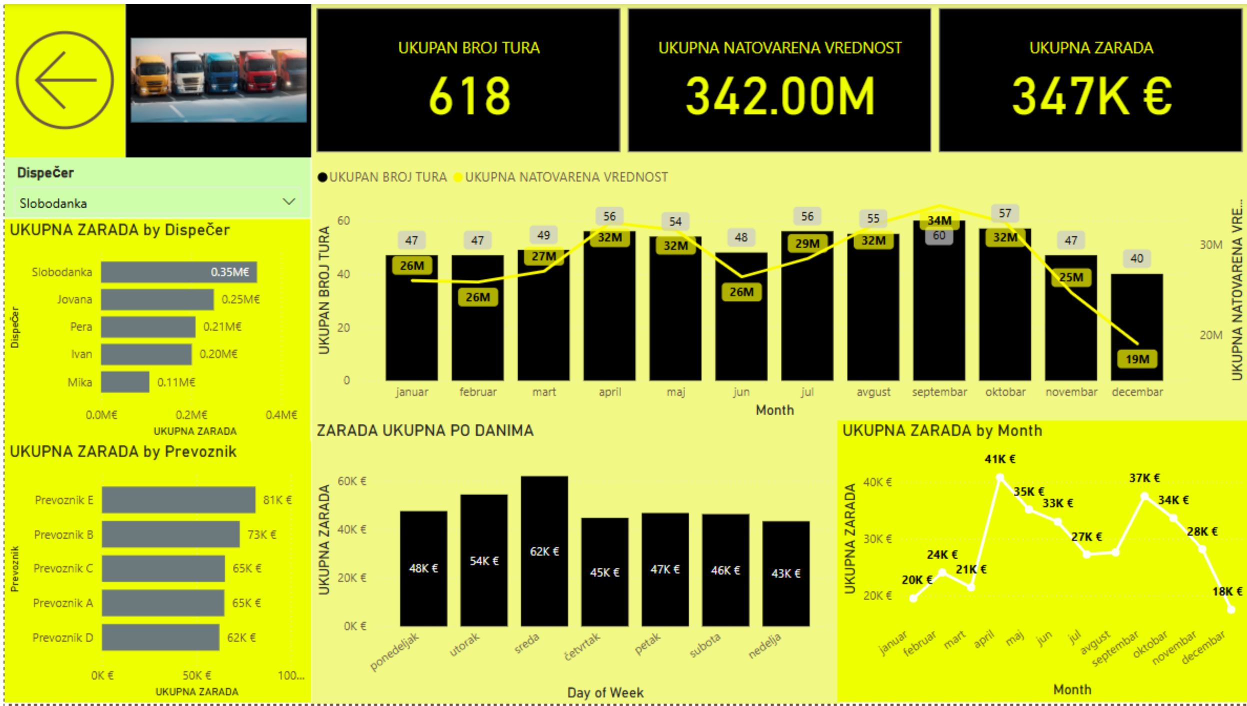
1. Dashboard Content

2. Total Revenue of Road Transport Agency

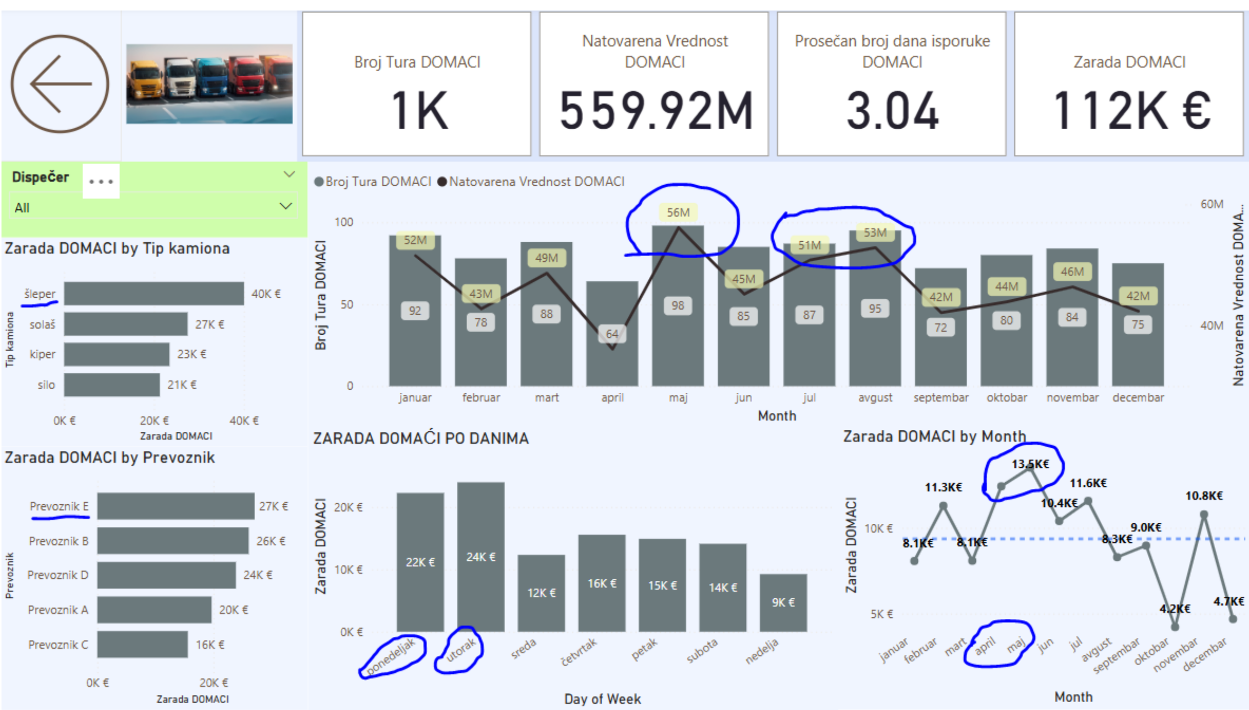
Analysis Summary:

- **Peak Loading Period**: The highest number of loaded trucks occurred from May to August, with a significant growth observed from April to May, particularly in the value of loaded goods. After August, the numbers gradually decreased. This trend aligns with the pattern seen in **total provisions**, which also saw a rapid increase from April to May and then a steady decline after August.
- **Provision Analysis**: Throughout the year, the transport agency recorded the highest provisions on **Mondays** and **Tuesdays**, while **Wednesdays** were the least productive days in terms of provision values.
- **Employee Performance**: Among the employees, **Slobodanka** achieved the best performance, while **Mika** had the weakest results.
- **Key Transport Partner**: The most important transport partner across the year was **Partner E**.
- **May to August Period Insights**:
  - During the period from **May to August**, the most productive days for provisions were **Sundays** and **Mondays**.
  - The most significant partners during this time were **E**, **C**, and **A**.

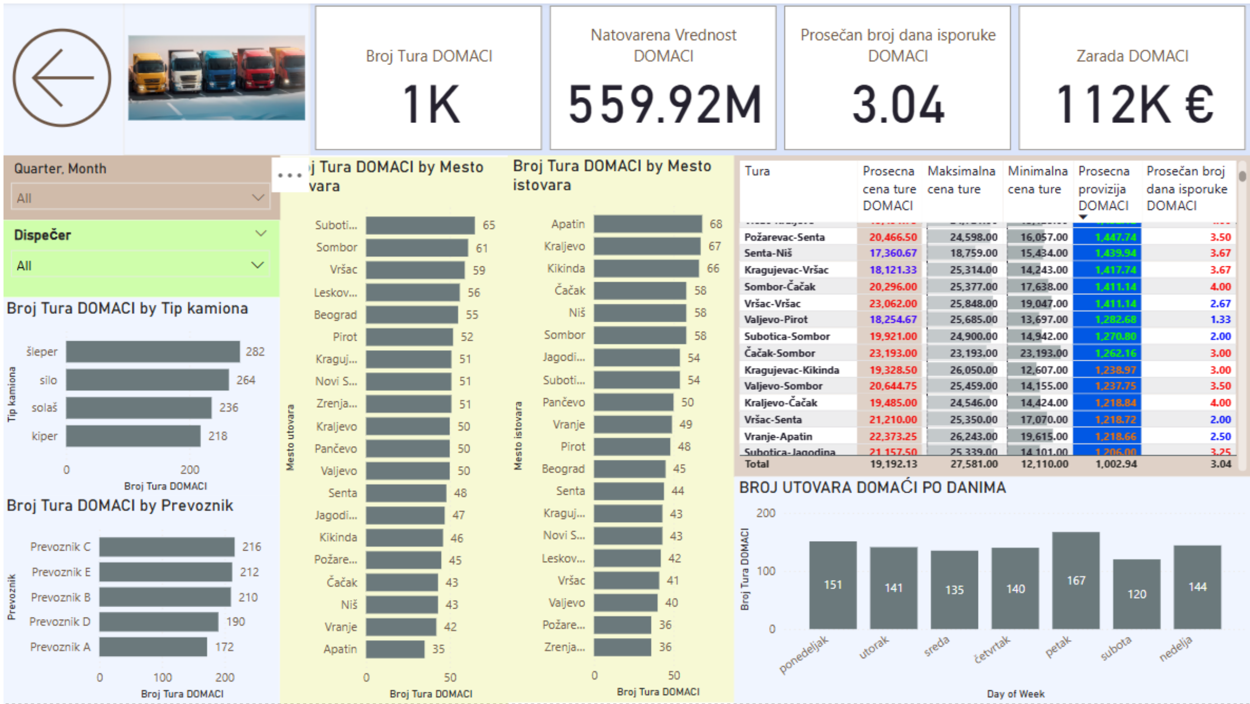




3. Total Revenue of Domestic Road Transport



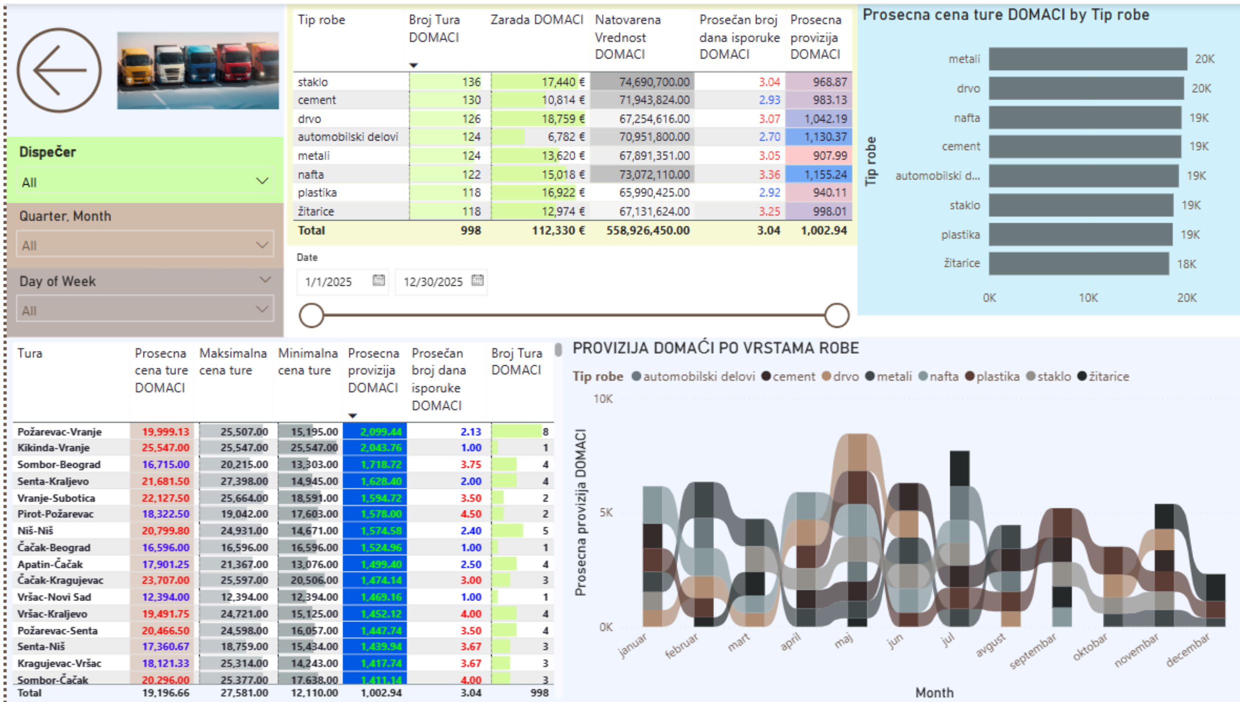
4. Loading and Unloading Locations in Domestic Transport



Route and Location Analysis:

- **Most Active Locations**: The highest number of loadings occurred in **Subotica**, while the most unloading locations were in **Apatin**.
- **Route Performance**: The data table clearly highlights which routes may be underperforming or questionable for the road transport agency's business. It also helps identify routes that require more attention or should be utilized more frequently to optimize overall efficiency.

5. Types of Loaded Goods in Domestic Transport

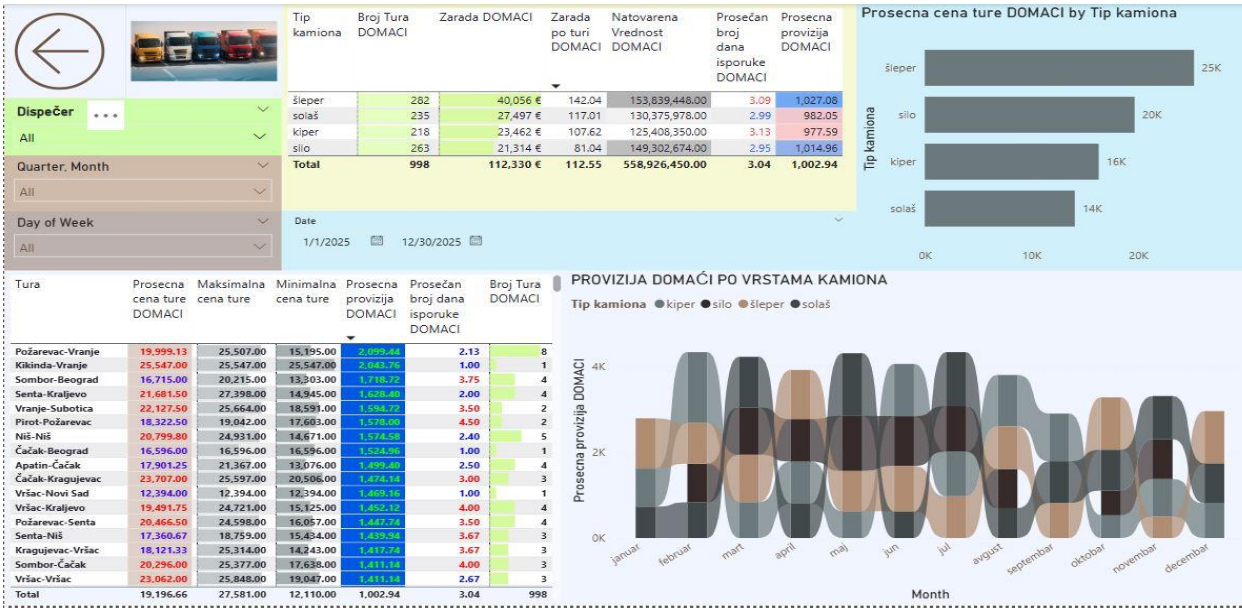


Tip robe	Broj Tura DOMACI	Zarada DOMACI	Natovarena Vrednost DOMACI	Prosečan broj dana isporuke DOMACI	Prosecna provizija DOMACI
staklo	136	17,440 €	74,690,700.00	3.04	968.87
cement	130	10,814 €	71,943,824.00	2.93	983.13
drvo	126	18,759 €	67,254,616.00	3.07	1,042.19
automobilski delovi	124	6,782 €	70,951,800.00	2.70	1,130.37
metali	124	13,620 €	67,891,351.00	3.05	907.99
nafta	122	15,018 €	73,072,110.00	3.36	1,155.24
plastika	118	16,922 €	65,990,425.00	2.92	940.11
žitarice	118	12,974 €	67,131,624.00	3.25	998.01
Total	**998**	**112,330 €**	**558,926,450.00**	**3.04**	**1,002.94**

Tura	Prosecna cena ture DOMACI	Maksimalna cena ture	Minimalna cena ture	Prosecna provizija DOMACI	Prosečan broj dana isporuke DOMACI	Broj Tura DOMACI
Požarevac-Vranje	19,999.13	25,507.00	15,195.00	2,099.44	2.13	8
Kikinda-Vranje	25,547.00	25,547.00	25,547.00	2,043.76	1.00	1
Sombor-Beograd	16,715.00	20,215.00	13,303.00	1,718.72	3.75	4
Senta-Kraljevo	21,681.50	27,398.00	14,945.00	1,628.40	2.00	4
Vranje-Subotica	22,127.50	25,664.00	18,591.00	1,594.72	3.50	2
Pirot-Požarevac	18,322.50	19,042.00	17,603.00	1,578.00	4.50	2
Niš-Niš	20,799.80	24,931.00	14,671.00	1,574.58	2.40	5
Čačak-Beograd	16,596.00	16,596.00	16,596.00	1,524.96	1.00	1
Apatin-Čačak	17,901.25	21,367.00	13,076.00	1,499.40	2.50	4
Čačak-Kragujevac	23,707.00	25,597.00	20,506.00	1,474.14	3.00	3
Vršac-Novi Sad	12,394.00	12,394.00	12,394.00	1,469.16	1.00	1
Vršac-Kraljevo	19,491.75	24,721.00	15,125.00	1,452.12	4.00	4
Požarevac-Senta	20,466.50	24,598.00	16,057.00	1,447.74	3.50	4
Senta-Niš	17,360.67	18,759.00	15,434.00	1,439.94	3.67	3
Kragujevac-Vršac	18,121.33	25,314.00	14,243.00	1,417.74	3.67	3
Sombor-Čačak	20,296.00	25,377.00	17,638.00	1,411.14	4.00	3
Total	19,196.66	27,581.00	12,110.00	1,002.94	3.04	998

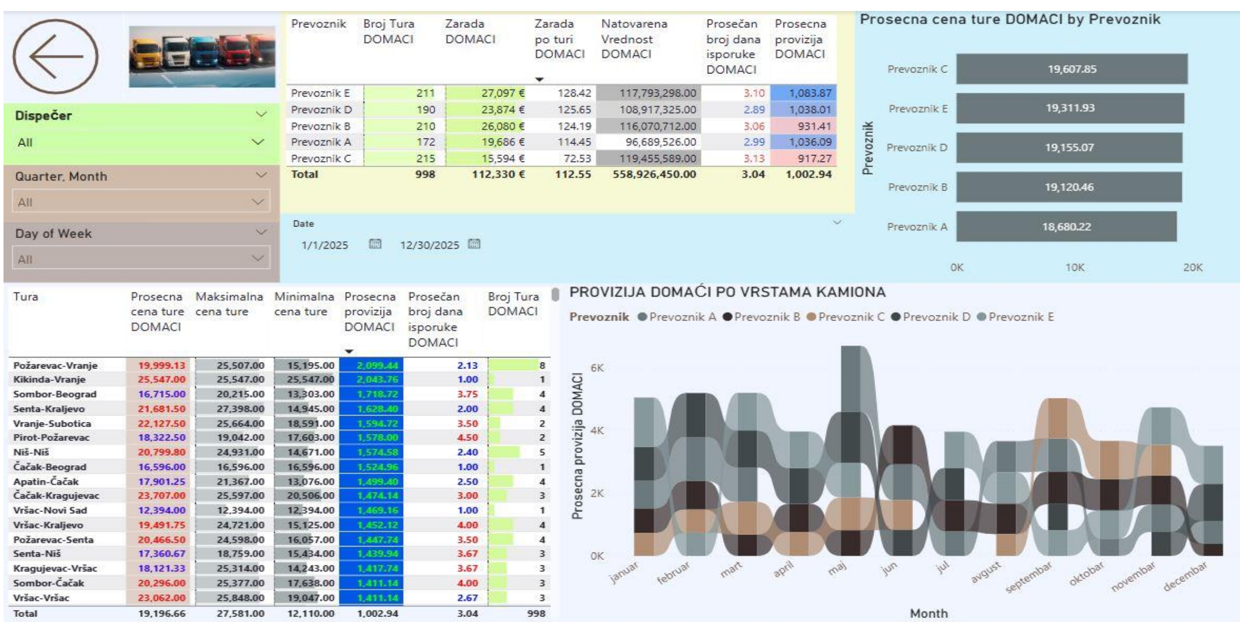
This section of the dashboard is designed to optimize the number of trips for transporting specific types of goods along defined routes. It provides insights into the average number of days from loading to unloading, as well as the average loading cost, helping to increase or decrease trip frequency based on efficiency and cost-effectiveness. The dashboard also reveals which seasons generate the highest total commission for transporting specific types of goods, allowing for better strategic planning and resource allocation throughout the year.

6. Types of Trucks in Domestic Transport



Same but for type of trucks.

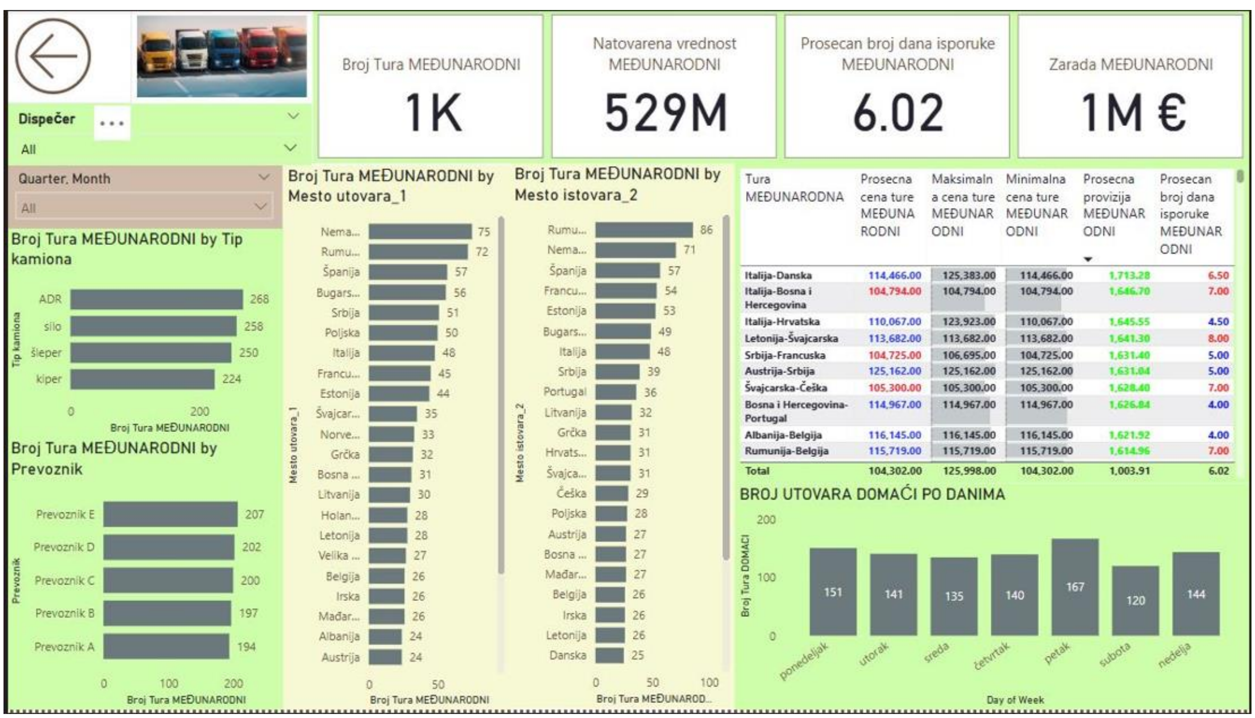
7. Partners in Domestic Transport



Same by partners.

8. Total Revenue of International Road Transport

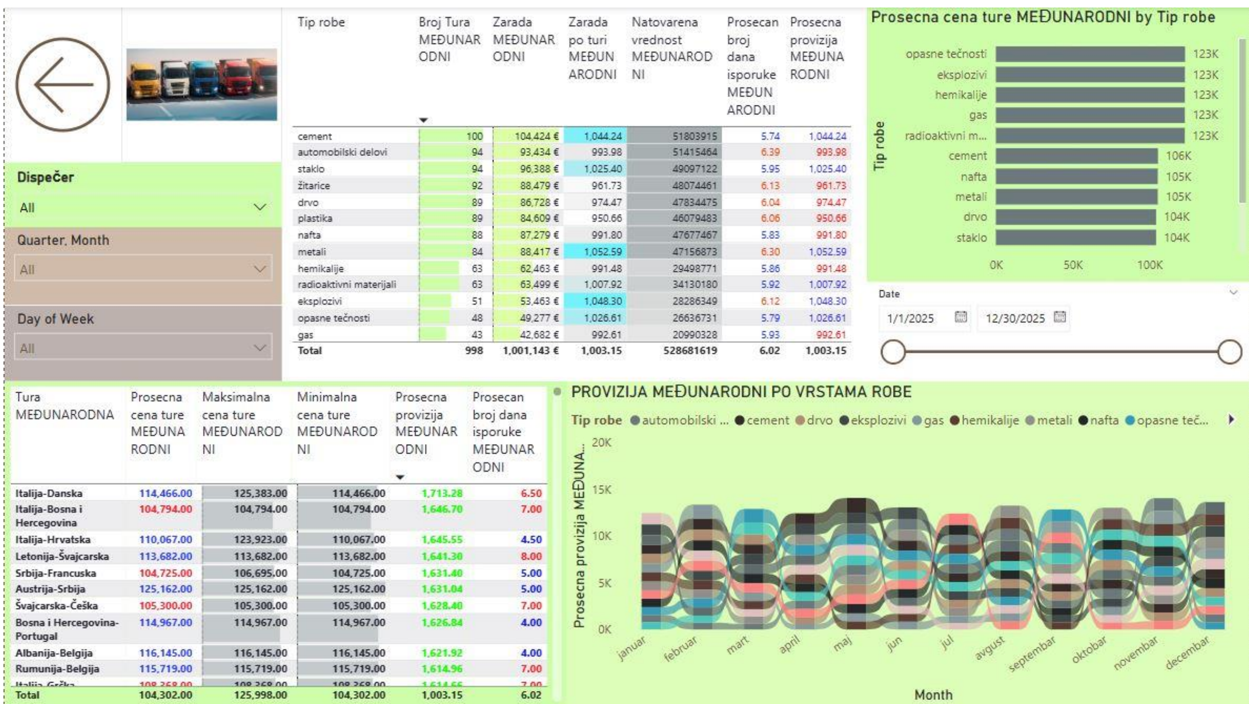
9. Loading and Unloading Locations in International Transport



Route and Location Analysis:

- **Most Active Locations**: The highest number of loadings occurred in **Germany**, while the most unloading locations were in **Romania**.
- **Route Performance**: The data table clearly highlights which routes may be underperforming or questionable for the road transport agency's business. It also helps identify routes that require more attention or should be utilized more frequently to optimize overall efficiency.

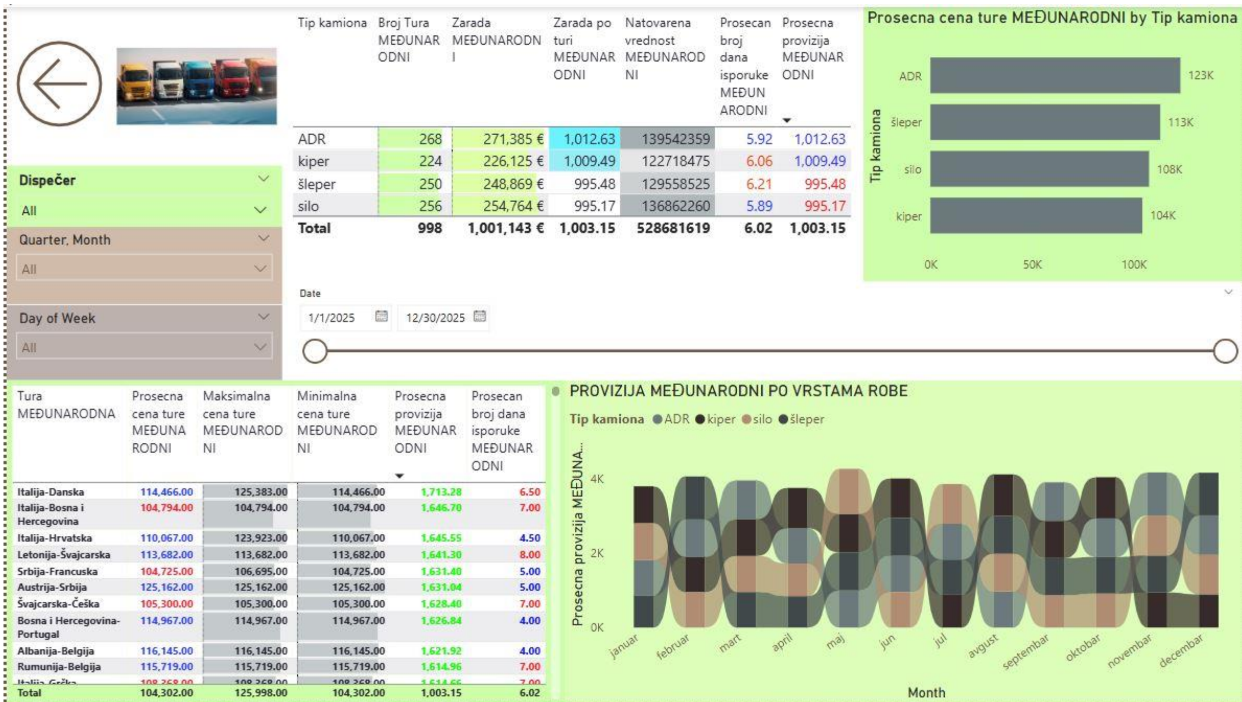
10. Types of Loaded Goods in International Transport



Key Insights:

- The highest revenue in international road transport was generated from the transportation of cement, followed by automotive parts and glass.
- A major issue in the transportation of automotive parts is the high average delivery time combined with a low average commission, indicating inefficiency in this segment.
- Increasing the number of trips for transporting hazardous liquids could be a beneficial move, as this segment shows potential for growth.

11. Types of Trucks in International Transport



ADR trucks dominate in terms of engagement, which is a positive trend due to their below-average delivery times and the highest average commission per trip.

12. Partners in International Transport